



Smart Expense Splitter and Group Budget Management System

Team ID: [__3__] | Section: [4_] | Members: [Challa Amrutha – 2520030412.], [Dhurga Meenakshi Yenduri – 2520030270.], [Earla Ashwith Reddy – 2520030475.] | Guide: [Dr. Yerragudipadu Subbarayudu]
Database Systems Engineering and Distributed Backend Development (25CS1302E) | KLH Bachupally Campus, CSE | PBL 2026-27

Abstract

- A web system for friends, roommates and travel groups to record shared expenses, split them flexibly and track a group budget, built with Node.js, Express and PostgreSQL.
- It warns early about budget overruns and shows the fewest payments needed to settle up.

Problem Statement & Objectives

- Problem: Shared expenses are tracked in chats and notes. Most apps only split equally and show what has already happened, so overspending is noticed late.
- Objective 1: Flexible splits (equal, percentage, exact)
- Objective 2: Predict overruns and test what-if cuts
- Objective 3: Settle up with the fewest payments
- Scope: Friends, roommates and travel groups; groups, members, expenses, budget, analytics

Existing vs Proposed System

Feature	Existing System	Proposed System
Splitting	Mostly equal splits	Equal, percentage or exact amount
Budget	Totals after the fact	Month-end projection and alerts
Insights	Plain expense list	Category charts, unusual-expense flag
Settling up	Many separate payments	Fewest payments for all balances
Limitation / Gap	No early warning or what-if planning	What-if cuts across several categories

Feasibility Study

- Technical: Common, well-documented tools (HTML, CSS, JavaScript, Node.js, SQL); runs on a normal laptop.
- Economic: Open-source stack (Node.js, PostgreSQL, Chart.js), so no licence cost; free-tier hosting is possible.
- Operational: Simple dashboard with forms and charts; one database keeps maintenance easy.
- Schedule: Design, database, backend, frontend and testing fit within the semester.

Results & Testing

- CRUD works for groups, members and expenses
- Bad splits and inputs are rejected
- Settlements match hand-checked examples

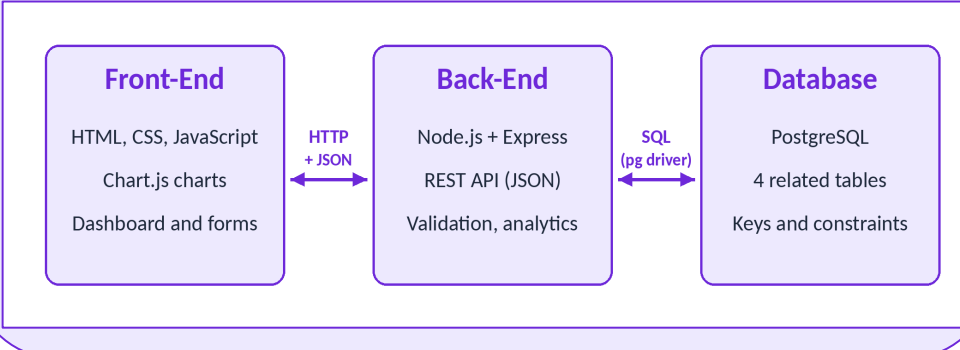
Conclusion

- A working expense splitter with budget prediction on PostgreSQL
- All three objectives met: splits, prediction, settlement

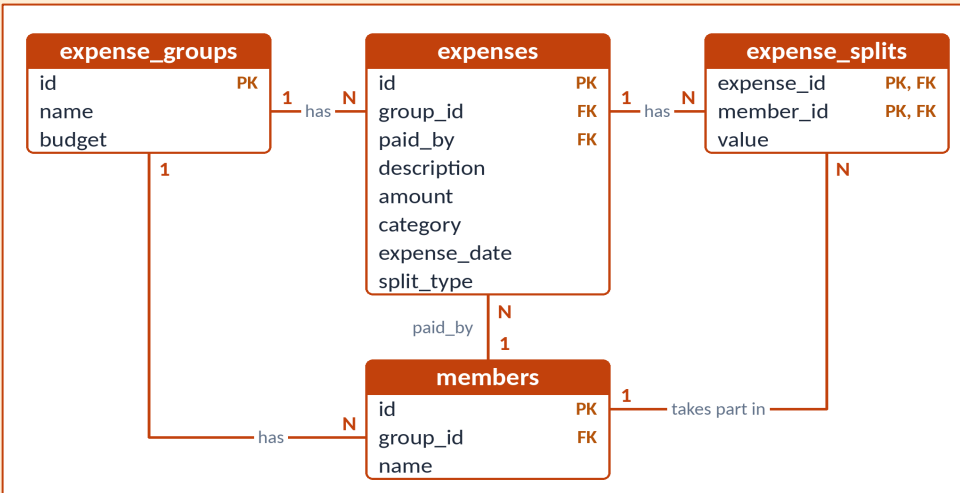
Tools & Technologies

- Front-End: HTML, CSS, JavaScript, Chart.js (charts)
- Back-End: Node.js + Express, REST APIs (JSON)
- Database: PostgreSQL (SQL, pg driver)
- Tools: VS Code, Git & GitHub, npm, dotenv, psql

System Architecture



ER Diagram & Database Design



Implementation Screenshots

